Selecting Periodic Domains in Cellular Automaton Spacetime with Bernoulli NML

Joshua Herman[1]
[1]Affiliation to be added

Motivation. Cellular automata often form broad periodic domains together with defects, particles, and interfaces. Those domains are visually obvious in many rules, but choosing *which* global period best explains a finite spacetime is less obvious. We propose a fast selector for that task. The key move is to treat each candidate period/shift pair as a model class, fit the nearest periodic background by majority vote on its orbit classes, and then score the candidate with a Bernoulli normalized maximum likelihood (NML) criterion.

Method. For a candidate period p and shift $\mathbf{s}$, the translation $(t, \mathbf{x}) \mapsto (t + p, \mathbf{x} + \mathbf{s})$ partitions spacetime into orbit classes. Each orbit contributes one background bit, chosen by majority vote, so fitting is linear in the number of observed sites. The NML score combines Bernoulli data fit and a complexity penalty. That penalty is essential because along velocity-matched divisibility chains, larger templates can only improve raw residual and raw likelihood.

Empirical picture. Across the current 1D, 2D, and 3D case studies, the selected period either stabilizes immediately or changes only a small number of times before locking. Rule 54 stabilizes at period 4 from the first measured horizon; Rule 110 passes through a short transient before locking at period 4 under the period-first selector; S24/B11 in 2D later shifts from period 1 to period 2; the tested 3D diamoeba rule remains at period 1 throughout the observed range. The winning margin grows after locking, which matches the model-selection picture behind the full paper. The penalty also changes the selector qualitatively: on 105 nontrivial Life-like rules at $T = 100$, Bernoulli NLL without the penalty chooses the largest tested period in every case, whereas Bernoulli NML selects period 1 for 97 rules, period 2 for 7, and period 8 for 1.

Why this is useful for ALIFE. The method is deliberately modest. It does not replace local causal-state reconstruction, and it does not claim that the selected period is always a physicist's "true" background period. Instead it provides a global front end: a compact domain hypothesis, a residual mask, and a complexity-penalized way to compare periodic explanations across many rules. That makes it suitable for large surveys, for visually driven poster presentation, and for follow-on analysis of particles, defects, and coherent structures.

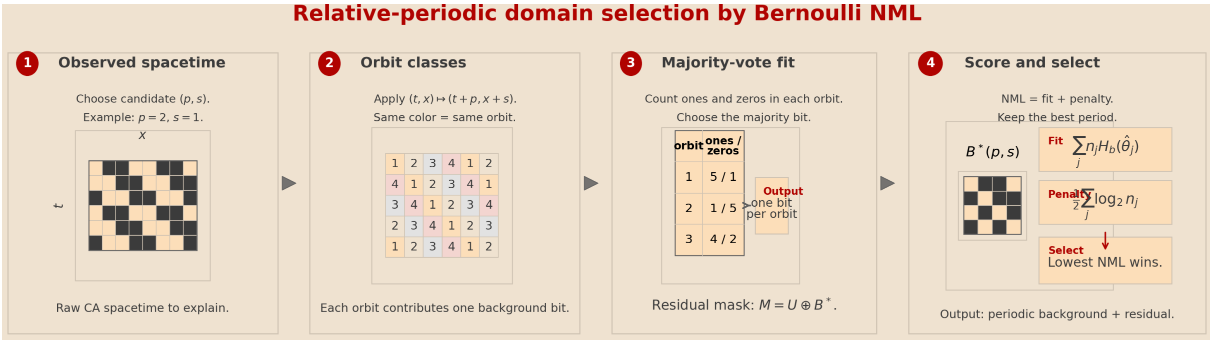

Figure 1: Algorithm sketch. A candidate period/shift pair induces an orbit partition of spacetime. Majority vote on each orbit gives the nearest periodic background $B^*(p, \mathbf{s})$; the residual mask is $M = U \oplus B^*$. Candidates are scored by $\mathrm{NML}(p, \mathbf{s}) = \sum_j n_j H_b(\hat{\theta}_j) + \frac{1}{2}\sum_j \log_2 n_j$, and the selected period is $p^* = \arg\min_p \min_{\mathbf{s}} \mathrm{NML}(p, \mathbf{s})$.

Figure 2: Representative stabilization patterns. The top row shows selected period versus horizon; the bottom row shows the winning margin in bits. The period either stabilizes immediately (ECA-54, diamoeba3d) or changes only a small number of times before locking (ECA-110, S24/B11).

References

Crutchfield, J. P. and Hanson, J. E. (1997). Computational mechanics of cellular automata: An example. *Physica D*, 103:169–189.

Gr"unwald, P. (2007). *The Minimum Description Length Principle*. MIT Press.

Hanson, J. E. and Crutchfield, J. P. (1992). The attractor-basin portrait of a cellular automaton. *Journal of Statistical Physics*, 66:1415–1462.

Rupe, A. and Crutchfield, J. P. (2018). Local causal states and discrete coherent structures. *Chaos*, 28:075312.

Shalizi, C. R., Haslinger, R., Rouquier, J.-B., Klinkner, K. L., and Moore, C. (2006). Automatic filters for the detection of coherent structure in spatiotemporal systems. *Physical Review E*, 73:036104.

Shtarkov, Y. (1987). Universal sequential coding of single messages. *Problems of Information Transmission*, 23:3–17.